SKILL CLIMAX V1.0

Problem Statement ID – 001

Problem Statement Title-Predictive Logistics AI

Theme- AI For Smart Urban Technologies

Team ID- To Be Updated

Team Name - NeuroNova

IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

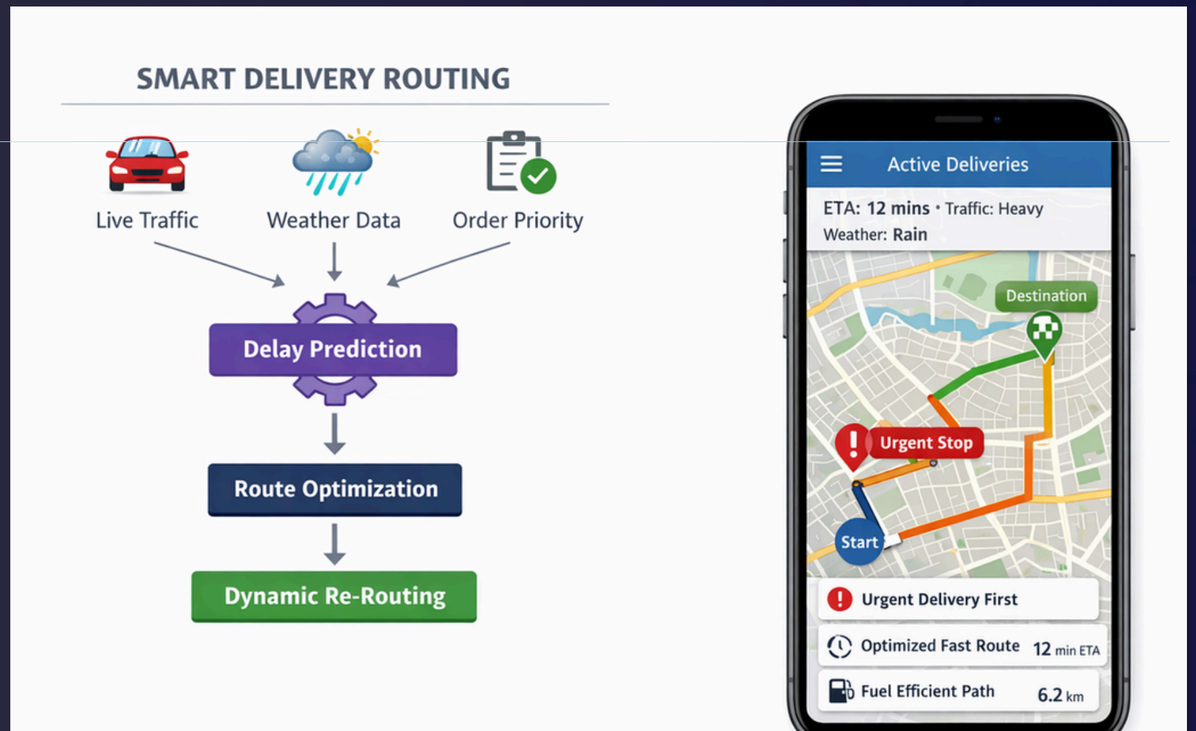
Detailed explanation of the proposed solution-We propose an DRL(Deep Reinforcement Leaning)based system that dynamically updates last-mile delivery routes using real-time traffic, weather, and delivery priority data.

How it addresses the problem- Updates routes in real time to save time and fuel.

Innovation and uniqueness of the solution - Routes that adjust instantly,Smart priority handling, Built for real city traffic.

TECHNICAL APPROACH

- Technologies to be used - Python(Core), TomTom(For Traffic Data), Google OR tools(Routing), XG Boost(Service Time prediction).
- Methodology and process :



FEASIBILITY AND VIABILITY

- Analysis of the feasibility of the idea:- Can be built with Python and ML , Uses real-time traffic, weather, and delivery info,Works for multiple vehicles in a city.
- Potential challenges and risks:- Needs fast updates,Events or Accidents can cause delays, Inaccurate weather/ Traffic.

CHALLENGES

- Strategies for overcoming these challenges -

  Inaccurate Weather/ Traffic - Use multiple sources and validate updates.
  Events / Accidents - Re-route dynamically in real time.
  Need Faster Updates - Use fast algorithms and cloud updates.

IMPACT AND BENEFITS

- Potential impact on the target audience:-

  Customers: Faster, on-time deliveries.

  Delivery teams: Easier route planning, less stress.

  Companies: Lower fuel costs, better efficiency.

- Benefits of the solution (social, economic, environmental, etc.) :- Happy Customers , Less Pollution& Costs , Lower Fuel Usage.

RESEARCH

Details / Links of the reference and research work-

Machine Learning-Enhanced Last-Mile Delivery (ML-CALMO): https://www.mdpi.com/2076-3417/15/21/11320

Real-Time Traffic Prediction & Reinforcement Learning for Delivery : https://link.springer.com/article/10.1007/s44327-025-00112-z